

Hacettepe University | Computer Engineering CMP 720 EMBEDDED SYSTEMS DESIGN

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Midterm Project Peer Evaluation Form

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Instructions

For each project, briefly summarize the work and provide your evaluation **individually (not as a group)**. Your response should reflect your understanding of the problem, methodology, and evaluation plan, and include critical insights, strengths, limitations, and possible improvements. Skip your own group's evaluation and simply state that it is your group.

Students are required to evaluate only the projects presented within their own class group (Hacettepe or ASELSAN).

Guiding points (address in one coherent paragraph):

- What problem is being solved, why is it important, and what are the most important embedded system design considerations for this problem (e.g., timing, reliability, energy, safety, security, scalability, etc.)?
- What is the main idea of the proposed methodology?
- What do you think are the strengths and limitations of the approach?
- Is there anything you would improve or do differently?
- What is the most critical or challenging component?
- Is the evaluation plan sufficient? What, if anything, is missing?

Group 1

Umut Kalman, Eren Karadağ

Project Title: *Context-Aware Deadline Scheduling Under Memory Contention for Embedded Real-Time Systems*

This project aims to solve the problem of memory contention in RTOS applications. Most RTOS schedulers does not deal with how much memory a task will require and how it will compete with other tasks memory-wise running at the same time. Therefore, in most RTOS applications undesired jitters, delays or even exceptions due to this memory contention. This problem requires a solution which emphasizes timing and reliability as the solution to this problem should make the system faster by eliminating jitter while not causing a critical system failure. The main idea of this project also focuses on the timing and reliability improvements in the system. The proposed solution uses the standard EDF (Earliest Deadline First) as the baseline scheduler and add a memory aware scheduler algorithm of their own to it, which is written in the user space. Since the main scheduler is still an industry standard one and they mainly wrap their algorithm with it in the user space, solution seems to be a reliable one. However, in terms of timing I am not sure that an algorithm written in the user space in order to do task scheduling would save time, since in the proposed problem it is said that this solution is to reduce the jitter and delay caused by tasks competing on the same memory. If the memory competition isn't very high but the algorithm still sees them as contenders it will make a computation that will take time and maybe this time will be greater than the memory competition caused jitter, resulting in a slower system. The proposed solution also takes this probability into account and they propose an online slack proxy where the system does not try to optimize when the slack is high enough. However, I didn't understand how the optimal slack value is determined as the jitter amount caused by this memory competition is not something that is clear. There could be a sweet spot where this computation causes more delay in comparison to the jitter. Maybe a formulation regarding the memory contention vs jitter would make the proposed solution more stable. Therefore, I believe timing is the most critical part of this solution. The evaluation plan for this project sounds sufficient for the most part as the generated optimization result is compared with a known optimization algorithm ACO (Ant Colony Optimization) offline, and the timing measurements will done in a real embedded device where a valid real time operating system FreeRTOS is used. Maybe testing the solution on different boards and on different operating systems would help in robustness analysis as different "ARM Cortex" series such as "A", "R" and "M" have different behaviors in applications like these. Maybe the R series would prove sufficient to not need such an algorithm.

Group 2

Yusuf Çağrı Öğüt

Project Title: *Hybrid Cellular-LoRa Network Based Agriculture Monitoring and Control Application*

This project aims to solve the remote maintenance problem of the hobby gardens in rural areas where there is either a very limited internet connection or none. The solution to this problem must emphasize reliability, energy and safety as it is important to ensure that the garden is not maintained poorly, the installed control system is safe for its environment and it consumes low energy. The low energy consumption is also emphasized multiple times in the proposed solution where the formulation and numerical data almost entirely about the energy consumption. The main idea of the proposed methodology is communicating with an embedded computer inside a hobby garden house via sending an SMS to it and making that embedded system send operation signals to the control system on the garden itself via LoRa, a long range low power small band-width communication protocol. The proposed solution to this problem makes sense because the internet connection is a real problem in the rural areas and communicating with a remote system via SMS is a valid option. The LoRa communication is also meaningful however it has low bandwidth and if

the system require more data transfer for logging monitoring it may be unefficient. There are other methods such as WiFi HaLow which is also a WAN and it has much higher bandwidth. However, if the required bandwidth is not high than since LoRa is cheaper it is logical. The system may suffer some reliability problems if the control system is not designed fault-proof. Since there are sensor and actuator communications, such as commanding an actuator for opening or closing the water canal valve, the fault on executing the desired command may be dangerous for the crops. I would suggest some watchdog timers preventing system collapse or some master slave power switching mechanisms so that the master device can reset the slave devices in timeout states. Also to ensure the systems most critical constraint, the low power consumption, it would most certainly help to analyze the embedded boards power consumptions in detail. For example some boards have different pin configurations where the pins (GPIO) are set to analog state, which ensures that there is no source power drawn to those pins. Maybe implementing such configurations help to obey the proposed 0.10 (approximately) mAh per day constraint. The evaluation part of the system makes sense as the evaluation includes nearly all of the system components and the real environmental conditions.

Group 3

Yusuf Tahir Orhan

Project Title: *Implementation of DeFT: A Deadlock-Free and Fault-Tolerant Routing Algorithm*

This project aims to solve the routing problem in 2.5D chiplet based NoC (Network on Chip) systems. The routing of such systems is very important because poorly designed routes may arise some deadlocks caused by the dependencies between nodes and if the VL's (Vertical Links) fails it may result in a total system failure. The solution to such a problem has a lot of system considerations in reliability and timing since the system must tolerate the VL faults and routing decisions must result in a deadlock-free fast system in order to operate desirable. The approach proposed in the solution consists of two parts: deadlock-free routing with VN's (Virtual Network) and dynamic VL selection. The virtual networks partition the traffic of the system into multiple logical networks and tries to ensure there are no cyclic dependencies. Dynamic VL selection will make the system follow a working VL path in order to achieve fault tolerance. The solution includes a strong architecture having deadlock freedom, VL fault tolerance, adaptive routing to increase performance. However, its VN assignment rules may decrease the performance as it is too focused on deadlock-free rule and the VN management complexity may increase too much with the system size. It is a very difficult problem and any proposed solution is mostly a meta-heuristic approach, so it may help to implement an online training machine learning methodology, which are used for a lot of problems that require meta-heuristic approaches, for VL selection and VN partitioning. The most critical part of the proposed solution is the VL fault tolerance in my opinion, as the system would redefine entire routes on such cases, ensuring the deadlock-free constraint is a very hard problem in situations like this. The evaluation plan is mostly sufficient in my opinion as it uses Noxim, an accurate simulation environment for applications like these, and it uses different traffic models and a stochastic VL fault mechanism. However, even though the price of such systems is high, if it was used on a real NoC system it would be better.

Group 4

Mehmet Yusuf Sezgi, Rana Elif Albayrak

Project Title: *Real-Time CAN Decoder*

This project aims to implement a real-time CAN decoding system which is to decode raw CAN bus data in real time on an embedded system. The system will have high traffic rate (around 300 us per frame), limited CPU and memory and no DBC (Database CAN) file which means there will be no prior definition of the signals. This project is important because it enables diagnostics and

monitoring in an unknown CAN bus system and it can be used for reverse engineering purposes. The design considerations for this system are reliability, time and security. The system should manage the no frame loss constraint and while doing this it should also obey the heavy time constraints and it shouldn't intervene with the integrity of the CAN bus, it should act as a passive node on the bus to maintain security of the system. The methodology of the solution is that reading can bus data via SPI, which has the highest priority, then passing the raw data to queue and then process the data from queue to decode the information out of it, medium priority, then passing the decoded data to UART and SD card logging, low priority. The strengths of this system are that it has explicit handling of timing, memory management and SPI bus contention handling, which arise from CAN bus data acquisition and SD card logging requiring the same bus. However, the system has limitations regarding the decoding as there is not a clear accuracy measurement as reverse engineering without the DBC file is a fragile method and the proposed system does not contribute to the decoding method enough. The proposed solution includes more systems engineering than algorithmic contribution. The proposed methodology could use some clarification on decoding algorithm (light pre trained ML model such as the proposed prior work "CANClassify") and it may also benefit from some other logging system that does not use SPI interface since the CAN bus reader is the most critical component on the board and its contender is a mere logger. Most challenging parts of the project are decoding the raw data from the queue before it fills up with the new data resulting in frame loss and managing the SPI contention between SD logging and CAN frame acquisition which can also result in frame loss. The evaluation plan is somewhat solid because it has good test scenarios where there is a high bus load and it uses real-world data collection. However, in my opinion, there is not enough emphasis on the latency of the decoding algorithm and how it will be so fast that the messages will be decoded and logged before the queue fills up with the new data. The evaluation plan would be much more solid one with an analysis regarding these.

Group 5

Metin Irkıcatal, Anıl Arda

Project Title: *Edge-AI Based Smart Access Control System*

ASELSAN group

Group 6

Oğulcan Uğuroğlu, Barış Büyükyılmaz

Project Title: *Contention-Aware Duplication Scheduling for NoC Systems*

This project aims to improve the DAG (Directed Acyclic Graph) task scheduling on MpSoC and NoC systems. The traditional approaches don't cover the network contention, bandwidth limit and routing conflicts that may be present in the system especially if it is a NoC architecture and this is the scope of the proposed solution. There are many NoC architectures present in the modern embedded world and the communication inside these systems cause bottleneck in most cases. This is why the proposed solution is important as it emphasizes on the co existing computing and communication constraints. The design considerations for this solution are timing, scalability and reliability as the system should obey the strict time constraints of such systems, should work for various sizes of NoC architectures and must be reliable in terms of fault tolerance and routing errors. The approach introduces an evaluation of the trade off computation and communication, which is to prevent the process of duplicating tasks blindly to avoid communication. This proposed solution implements a heuristic method that allows task duplication if the communication constraints is not a problem. This method addresses a known gap in classical scheduling models according to the limited research that I have done, and unlike them, it explicitly considers network contention. Also, this work targets communication-intensive workloads to fill the mentioned gap even more to prove that

the methodology is efficient. However, as most of the optimization focused solutions in this field, this is a heuristic approach and it is not guaranteed to result in an optimal scheduling. Also, there is a risk of some potential overhead when evaluating duplication trade-offs that may increase scheduling time since it is a heuristic solution. Also, the system follows a static scheduling model, which means that all scheduling decisions are made offline. This situation makes the problem a lot easier but as an improvement to the system an online scheduling, adapting method would be a really good solution to such a problem however, I don't have the necessary knowledge to improve this system in such a drastic way. The system also could use some NoC routing algorithms as it emphasizes on these systems and it would certainly help to limit the duplication when there is a path fault. The most challenging part of this system is definitely the estimation of communication contention. Since the scheduling algorithm is offline this makes the wrong estimations about these communication contentions much more fatal. The evaluation plan is solid as the simulations they plan to make includes the necessary elements such a different CCR (communication-to-computation) ratios and various NoC configurations and it focuses on the makespan of the system as a result which makes sense. However, this work also lacks a hardware verification which is necessary to see the use case with real-world DAG workloads.

Group 7

Ahmet Yiğitalp Demirci

Project Title: *Real-Time Energy-Constrained UAV Optimization*

ASELSAN group

Group 8

Sami Enes Yılmaz, Kadir Emre Avcı

Project Title: *SmartAquaria System*

ASELSAN group

Group 9

Mehmethan Ayrım

Project Title: *Energy-Aware Mixed-Criticality Scheduling*

ASELSAN group

Group 10

İrem Gül Subaşı

Project Title: *Fault Injection and Resilience Profiling of Lightweight Transformers*

ASELSAN group

Group 11

Süleyman Ege Karakaya

Project Title: *Adaptive Data-Movement Strategy for Crypto Accelerators*

ASELSAN group

Group 12

Gökçer Erol

Project Title: *Acoustic Localization System for Search and Rescue*

This project aims to develop a distributed acoustic localization system to detect and locate survivors trapped under debris by analyzing sound signals and locate them via TDoA (Time Difference of Arrival) algorithm. This work is important because time is critical in disaster scenarios and traditional search methods are slow, and the proposed solution provides a fast localization method which is a high impact solution on saving lives. Proposed solution has timing, reliability and energy design considerations since, TDoA requires microsecond-level synchronization to be accurate, it must work on harsh environments including a lot of dust, debris and noise, and finally

it requires an efficient use of batteries to be mobile and be operable for long periods. The main methodology include distributed sound sensing from different nodes, synchronization of the timing info with GNSS-based pulse-per-second, localization via TDoA where nodes send the central processor timestamps of detected events and it computes the possible area of the survivor which is causing the sound. The communication between the central computer and the nodes are established by LoRa for long range communication and the filtering of the sound and acoustic event classification is done with lightweight ML models. The proposed methodology is strong because it has a real-world relevance, accurate time synchronization method (PPS), accurate localization approach (TDoA), low power consumption (LoRa) communication. However, its limitations are, unreliable GNSS signals in jamming or spoofing present areas, LoRa bandwidth limitations, sound attenuation and reflections in unpredictable environments. These problems may make the system inefficient in some environments. So some possible improvements are, instead of using only sound to detect survivors, some nodes that has thermal imaging cameras can be used in addition. Also, instead of depending only on GNSS maybe a network based synchronization can be used where there are networks present. The most critical parts of the system is definitely the precise time synchronization and reliable TDoA estimation since the localization directly depends on the timestamps and microsecond errors could result in meter-level localization errors. Clock drift and GNSS signal loss could create fatal errors. The evaluation plan of thee project mostly includes the necessary tests for the algorithm as it will demonstrate a working prototype and real time localization and visualization will be done but the possible errors that I stated before should also be considered since GNSS signals and system clocks cannot be trusted every time. Also the LoRa bandwidth is another problem that should be considered so maybe selecting some long range high bandwidth alternative such as WiFi HaLow would be good.

Group 13

Emre Doğan, Yavuz Selim Kızıltaş

Project Title: *Optimization of Edge AI Pipeline on TDA4VM*

ASELSAN group

Group 14

Mehmet Yusuf Bilge, Sezil Ağırbaş

Project Title: *Closed-Loop Vestibular Stimulation System*

This project aims to design a system that improves user balance via vestibular stimulation. This work is important as there are so many injuries caused by falling (mostly elderly people) and such a solution may be useful for preventing those injuries by making people wear a minimal embedded device kit. The main considerations for this system are safety, reliability and time. The system should not cause more imbalance while trying to provide a better balance, the control loop and sensing of the human body state (stable, falling, slipped etc.) should be reliable in order to provide what is promised and the timing of the actions are critical as it wouldn't help if the state that is requiring the stimulation output has already passed. The methodology includes an IMU sensing body-motion (posture, head movement, sudden accelerations, body sway) and a controller applying stimulation signals via Galvanic Vestibular Stimulation method which is about applying low-level electrical currents to the bones behind the ears of the person. The project is a strong example of an embedded wearable system and serves a great purpose about preventing injuries. However, there are some limitations about this system as incorrect stimulation may disorient users and increase fall risk and estimating the correct signal is not an easy task as human vestibular responses differ among people as humans are highly variable. Also, the system is non linear as human balance is complex and not easily modeled. The system is highly sensitive to latency and will require following strict time constraints. The most critical part of the system is the design of

the feedback control algorithm in my opinion because the system we are controlling is a non linear system and it is not time invariant which is one of the hardest cases in control theory, also, the stimulation profile changes according to the person so the problem becomes even more complex. The evaluation plan for this project may never be enough as the results may change from person to person and it should certainly test for failure scenarios where the most reasonable outcome is to not stimulate the person at all to prevent worse scenarios. Maybe instead of using a deterministic control algorithm a learning algorithm can be presented alongside the control algorithm and the system may learn how it should behave for the person in the first days of usage.

Group 15

Ayberk Aygün, Berkay Taşkın

Project Title: *Quadrotor Simulation on NVIDIA Jetson*

My own group

Group 16

Yasin Arslan, Süleyman Sertaç Üst

Project Title: *Real-Time Audio Pipeline Analysis on Heterogeneous SoCs*

ASELSAN group

Group 17

Oğuz Kağan Yağlıoğlu, Emre Eşef Kılıçdoğan

Project Title: *Predictive Maintenance of BLDC Motors via Multi-Sensor Fusion*

This project aims to develop a BLDC (Brushless DC Motor) fault prediction system which will report the possible faults that may happen in short-term according to the acquired data. This work is important because BLDC motors are critical for manufacturing but are prone to failures, so detecting their fault beforehand would help in order to reduce the effect of such faults in production lines. System considerations for the project are reliability, scalability and energy. False negative results would be costly as the production line would stop for nothing, should be applicable for many BLDC motors and they will have some innate differences and finally the power consumption of the customized sensor devices are important as they include batteries. The main methodology includes some custom BLDC mounted multi sensor electronics that include a minimal STM32 MCU and accelerometer, ultrasonic microscope and temperature sensor on it. This electronic device which is mounted on the BLDC motor sends data to the central computer in order to evaluate the gathered sensor data and this communication is done by LoRa. The central computer then will send the data into cloud in order to get the predictions about the system's future state. The methodology here is also summarized by the Edge-Fog Computing Architecture terminology which is a widely used methodology in IoT applications as running heavy models on small devices would result in some serious latencies. The project is strong in terms of identifying the problem and finding its root causes, it answers to the question "Why do motors fail?" well and it has realistic industrial assumptions. The limitations of the project are that this project is highly dependent on the datasets and this may affect system performance badly in a scalability perspective and high sampling rate and the LoRa communication does not work really well. The evaluation plan seems solid as they are planning to evaluate with different models and they provide step-by-step experimentation. Only thing that raises questions for me is the acquisition of the data of real faulty BLDC motors.

Group 18

Hüseyin Eren Doğan

Project Title: *EdgeRAG: Offline RAG on Embedded Systems*

ASELSAN group

Group 19

Ali Bölücü

Project Title: *Spectral-Seeded NSGA-II for Many-Core Systems*

This project aims to optimize the application mapping into nodes of a NoC architecture in order to optimize communication cost, latency, energy and thermal distribution which covers nearly all difficulties of working on such a system. This work is important because it proposes a unique solution to the optimization problem of widely used NoC systems in every aspect of it. The key considerations for this system are energy, timing, scalability and reliability. Since it tries to optimize the thermal state of the chip it should not find high energy consuming solutions such as communication-heavy mappings, the optimizer should find solutions which obeys the real-time constraints of the system, it should operate on different scale NoC architectures and it should provide reliable results which doesn't cause any deadlocks, thermal problems etc. The main idea of the solution is about implementing the NSGA-II (Non-dominated Sorting Genetic Algorithm II) to optimize the proposed 9x9 NoC architecture with around 4 applications and around 81 tasks. It mainly tries to cluster highly communicating tasks and maximize the distance as much as possible between unrelated active cores to avoid local heating. Also, this method does not initialize NSGA-II randomly as it feeds the algorithm with some initial cluster-aware, desirable solutions. As it would be much harder to start from a random point in an N-dimensional plot. The proposed solution is strong because it improves the NSGA-II method with educated initial solutions which reduces search space effectively and considers the communication structure explicitly and it is a generalizable methodology. Its limitations are, static mapping and routing may not prove very useful in more dynamic task architectures, it may have a very large computation time regarding the system size, the initial condition may push the system into the local maximums (or minimums) if not selected properly and it still converge into local values if the updating policy is strict. However, the proposed solution also considers random sets it is a matter of optimizing the ratio of educated and random input sets. As an improvement maybe machine learning methods can be used alongside the NSGA-II algorithm, even though the training time of such a model would be greater than the proposed optimizer solving it, it may be faster in the long run. Also, it would be better if a minimal online optimizer can be used with the heavy offline optimizer. The most challenging part is the core of the project which is to design good clusters that reflect the communication of the nodes optimally as this is how the proposed solution succeeds, without it the solution doesn't have much meaning. The evaluation plan of the project is solid because it uses a stable and widely used simulation environment regarding the optimization, PlatEMO v2.8 (Evolutionary multi-objective optimization platform), and it is analyzed thoroughly in the proposed environment. If the evaluation also include different scale systems and different ratio of educated and random inputs it would certainly reveal a lot about the strength of the solution.

Group 20

İlter Doğaç Dönmez

Project Title: *Autonomous Obstacle Avoidance Vehicle using LiDAR*

This project aims to design an autonomous obstacle avoidance vehicle using LiDAR in a simulation environment. The process of simulating and evaluating every aspect of a desired system is invaluable because it saves a lot of money and time, also it prevents a lot of risks. This is why simulation works are important. The key considerations for this work is time and reliability as the system will simulate a physical reality and the events happening inside this simulation must follow certain time periods and even if it is not real time it should obey the related interaction and time differences between events to be accurate, and the system should use reliable LiDAR and other actuator simulations in order to design a reliable embedded device simulation. The main methodology of the proposed solution is a full-system simulation + scenario testing but it is missing how

these scenarios would be generated in order to evolve into efficient tests. Strength of the project is it simulates the system as a whole which include also include the Raspberry Pi 3 board simulation as this will be useful if the project will be implemented in the real-life later on. The limitations of the project come from simulation realism. Renode will be enough for simulating the CPU alongside with the algorithm running on it but it is not designed for simulating sensors, actuators and noise that these components may have. LiDAR modeling should be very precise and close to real-life for a simulation project such as this to be applicable and coding it directly to produce some summarized data in order to run test cases may be inefficient. Some possible improvements would be gathering some real logs from a similar system and using them by altering some of the data, or coding a simulation dataset in detail and adding random noise into those data sets according to some probabilistic distribution such as Gaussian distribution as it is the most common. It would also be good to add some sensor failures and delayed data in the simulation to ensure fault tolerance of the algorithm. The most challenging part of the project is creating the reliable test simulations as I have mentioned before and also obeying the timing constraints. Without these two the proposed solution would have no value in terms of a real-life application. Hence, the evaluation plan would be sufficient if the points about real-life simulation are considered and it would be better if the evaluation tests include some real actuator/sensor logs. Also, if the simulation also included some dynamical model simulating the behavior of the car it would be much better as the simulated car may move impossibly if not constrained according to a real-life dynamics model.